QF 604
ECONOMETRICS OF FINANCIAL MARKETS

LECTURE 9

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LECTURE OUTLINE

Econometrics of Stochastic Processes in Derivatives Pricing

01 DIFFUSION PROCESS ESTIMATION

02 INFINITESIMAL GENERATORS

03 SIMULATION METHODS

04 JUMP DIFFUSION PROCESS

01 Diffusion Process Estimation

Exact Discretization of Geometric Brownian Motion

The price process satisfies the diffusion equation

$$dp_t = \mu p_t dt + \sigma p_t dW_t, \tag{12.1}$$

where (W_t) is a standard Brownian motion. Its dynamics depends on a drift parameter μ and a volatility parameter σ. The estimation of volatility is essential since it allows determination of the risk-neutral probability and consequently the prices of derivatives on (p_t) (see Chapter 11).

By applying Ito's formula, we deduce the diffusion equation satisfied by the logarithm of the price. We get

$$d \log p_t = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW_t. \tag{12.2}$$

We find by direct integration

$$\log p_t - \log p_{t-1} = \mu - \frac{\sigma^2}{2} + \sigma(W_t - W_{t-1}),$$

or, equivalently,

$$\Delta \log p_t = \mu - \frac{\sigma^2}{2} + \sigma \varepsilon_t, \tag{12.3}$$

where (ε_t) is a standard Gaussian white noise. Therefore, the discretized version of the geometric Brownian motion corresponds to a Gaussian random walk with drift for the price logarithm.

Also called the Lognormal Diffusion Process

01 Diffusion Process Estimation

Maximum Likelihood Estimation of GBM

The ML method (conditioned on the first observation p_0) provides the estimators of the mean and variance of $\Delta \log p_t$:

$$\hat{m}_T = \frac{1}{T}\sum_{t=1}^{T} \Delta \log p_t, \qquad \hat{s}_T^2 = \frac{1}{T}\sum_{t=1}^{T} (\Delta \log p_t - \hat{m}_T)^2. \tag{12.4}$$

The ML estimators of the drift and volatility parameters are derived from their relations to the mean and variance parameters. They are given by

$$\hat{\mu}_T = \hat{m}_T + \frac{\hat{\sigma}_T^2}{2}, \qquad \hat{\sigma}_T^2 = \hat{s}_T^2. \tag{12.5}$$

It is known that the asymptotic variances of $\hat{m}_T$ and $\hat{s}_T^2$ are such that

$$V_{\text{asy}}(\hat{m}_T) = \frac{\sigma^2}{T},$$

$$V_{\text{asy}}\left(\hat{s}_T^2\right) = \frac{2\sigma^4}{T},$$

$$Cov_{asy}\left(\hat{m}_T, \hat{s}_T^2\right) = 0.$$

01 Diffusion Process Estimation

Maximum Likelihood Estimation of GBM

We infer the asymptotic variance-covariance matrix of the pair $\hat{\mu}_T, \hat{\sigma}_T^2$:

$$V_{asy}\hat{\mu}_T = \frac{\sigma^2}{T} + \frac{\sigma^4}{2T},$$

$$V_{asy}\hat{\sigma}_T^2 = \frac{2\sigma^4}{T},$$

$$Cov_{asy}\left(\hat{\mu}_T, \hat{\sigma}_T^2\right) = \frac{\sigma^4}{T}.$$

We observe that the estimators of μ and σ^2 are correlated.

01 Diffusion Process Estimation

Ornstein-Uhlenbeck Process

Exact Discretization

The dynamics of the process is described by the stochastic differential equation

$$dy_t = (\Phi - \lambda y_t)dt + \sigma dW_t \tag{12.7}$$

$$= \lambda(\mu - y_t)dt + \sigma dW_t, \tag{12.8}$$

where (W_t) is a standard Brownian motion. This equation has a simple discrete time counterpart (see Section 11.2.2):

$$y_t = \mu[1 - \exp(-\lambda)] + \exp(-\lambda)y_{t-1} + \sigma\left(\frac{1 - \exp(-2\lambda)}{2\lambda}\right)^{1/2}\varepsilon_t, \tag{12.9}$$

where (ε_t) is a standardized Gaussian white noise. This equation corresponds to a Gaussian AR(1) (autoregressive process of order 1) representation for the $(y_t, t \in Z)$ process

01 Diffusion Process Estimation

Ornstein-Uhlenbeck Process

Maximum Likelihood Estimators

We can easily apply the ML method to the autoregressive model in (12.9). Let us first reparametrize the autoregressive representation as

$$y_t = \mu(1-\rho) + \rho y_{t-1} + \eta\varepsilon_t, \tag{12.10}$$

where $\rho = \exp - \lambda$.

The ML estimators of the parameters μ, ρ, and η are asymptotically independent and equivalent to

$$\begin{aligned}
\hat{\mu}_T &= \frac{1}{T}\sum_{t=1}^{T} y_t = \bar{y}_T, \\
\hat{\rho}_T &= \frac{1}{T}\sum_{t=1}^{T} (y_t - \bar{y}_T)(y_{t-1} - \bar{y}_T) \Big/ \frac{1}{T}\sum_{t=1}^{T} (y_t - \bar{y}_T)^2, \\
\hat{\eta}_T^2 &= \frac{1}{T}\sum_{t=1}^{T} \hat{\varepsilon}_t^2,
\end{aligned} \tag{12.11}$$

where the residuals are defined by $\hat{\varepsilon}_t = y_t - \bar{y}_T - \hat{\rho}_T(y_{t-1} - \bar{y}_T)$. Their asymptotic variances are given by

$$V_{\text{asy}}\hat{\mu}_T = \frac{\eta^2}{T(1-\rho^2)}, \qquad V_{\text{asy}}\hat{\rho}_T = \frac{1}{T}(1-\rho^2), \qquad V_{\text{asy}}\hat{\eta}_T = \frac{2\eta^4}{T}.$$

01 Diffusion Process Estimation

Ornstein-Uhlenbeck Process

From the ML estimators of the parameters μ, ρ, and η, we easily infer the ML estimators of the parameters of interest

$$\hat{\lambda}_T = -\log \hat{\rho}_T, \qquad \hat{\sigma}_T^2 = -\frac{2 \log \hat{\rho}_T}{1 - \hat{\rho}_T^2} \hat{\eta}_T^2. \tag{12.12}$$

Their asymptotic variances are derived by the δ-method. For instance, we get

$$\begin{aligned} V_{\text{asy}}(\hat{\lambda}_T) &= \left[\frac{\partial(-\log \rho)}{\partial \rho} \right]^2 V_{\text{asy}} \hat{\rho}_T, \\ &= \frac{1}{T} \frac{1 - \rho^2}{\rho^2}, \\ &= \frac{1}{T} \frac{1 - \exp - 2\lambda}{\exp - 2\lambda}. \end{aligned}$$

The estimators $\hat{\lambda}_T$ and $\hat{\sigma}_T^2$ are asymptotically correlated since they both depend on $\hat{\rho}_T$. This is a consequence of time aggregation.

01 Diffusion Process Estimation

Cox-Ingersoll-Ross Process

The CIR process is:

$$dr_t = \kappa\left(\theta - r_t\right)dt + \sigma\sqrt{r_t}dw_t$$

where in the original paper, r_t is a short rate process, and W_t is a standard Wiener process ~ N(0,t).

Note: The Laplace transform method solution of the CIR process used in the Princeton text is not complete as advanced techniques in Bessel processes are also required. See more detailed proof in "Laplace transforms of stochastic integrals and the pricing of Bermudan options" – Lars Palapies, Decision in Economics and Finance 2023, Springer.

An indirect method of solution is shown here, using the ? known fact that sum of n squares of independent normal $X'_{i,t}$ ~ N(μ_i ,1) r.v.'s is chi-square $\chi^2_{n,}$ with n degrees of freedom and non-centrality parameter $\lambda = \sum_{i=1}^{n} \mu_i^2$.

Let $X_{i,t}$ be a normal r.v. at t with SDE

$$dX_{i,t} = -\frac{1}{2}\kappa X_{i,t}dt + \frac{1}{2}\sigma dw_t$$

01 Diffusion Process Estimation

Cox-Ingersoll-Ross Process

$$dX_{i,t} + \frac{1}{2}\kappa X_{i,t}dt = \frac{1}{2}\sigma dw_t$$

$$e^{\kappa t/2}dX_{i,t} + \frac{1}{2}e^{\kappa t/2}\kappa X_{i,t}dt = \frac{1}{2}e^{\kappa t/2}\sigma dw_t$$

$$d\left(e^{\kappa t/2}X_{i,t}\right) = \frac{1}{2}e^{\kappa t/2}\sigma dw_t$$

$$\int_s^t d\left(e^{\kappa u/2}X_i u\right) = \frac{1}{2}\sigma\int_s^t e^{\kappa u/2}dw_u$$

$$e^{\kappa t/2}X_{i,t} - e^{\kappa s/2}X_{i,s} = \frac{1}{2}\sigma\int_s^t e^{\kappa u/2}dw_u$$

$$X_{i,t} = e^{-\kappa(t-s)/2}X_{i,s} + \frac{1}{2}\sigma\int_s^t e^{-\kappa(t-u)/2}dw_u$$

This is essentially an Ornstein-Uhlenbeck (O-U) process or the Vasicek process in interest rate theory.

Its mean and variance are:

$$E\left[X_{i,t} \mid X_{i,s}\right] = e^{-\kappa(t-s)/2}X_{i,s}$$

$$V\left[X_{i,t} \mid X_{i,s}\right] = V\left[\frac{\sigma}{2}\int_s^t e^{-\kappa(t-u)/2}dw_u\right] = \frac{\sigma^2}{4}\int_s^t e^{-\kappa(t-u)}du = \frac{\sigma^2}{4\kappa}\left(1 - e^{-\kappa(t-s)}\right)$$

Conditional on $X_{i,0}$, then mean of $X_{i,t} = e^{-\kappa t/2}X_{i,0}$ for $t > s \geq 0$, $Var(X_{i,t}) = \frac{\sigma^2}{4\kappa}(1 - e^{-\kappa t})$

01 Diffusion Process Estimation

Cox-Ingersoll-Ross Process

We show $r_t = \sum_{i=1}^{n} X_{i,t}^2$ is a CIR process where $X_{i,t} \sim N(\mu_i, var_i)$ and is i.i.d.

$$d\left(X_{i,t}^2\right) = \frac{\partial X_{i,t}^2}{\partial X_{i,t}} dX_{i,t} + \frac{1}{2}\frac{\partial X_{i,t}^2}{\partial X_{i,t}^2} dX_{i,t}^2$$

$$= 2X_{i,t}dX_{i,t} + \frac{1}{2}2dX_{i,t}^2$$

$$= 2X_{i,t}\left(-\frac{1}{2}\kappa X_{i,t}dt + \frac{1}{2}\sigma dw_t\right) + \frac{\sigma^2}{4}dt$$

$$= \left(-\kappa X_{i,t}^2 + \frac{\sigma^2}{4}\right)dt + \sigma X_{i,t}dw_t$$

$$dr_t = dX_{1,t}^2 + dX_{2,t}^2 + dX_{3,t}^2 + \ldots + dX_{n,t}^2 = \sum_{i=1}^{n} dX_{i,t}^2$$

$$= \sum_{i=1}^{n}\left(-\kappa X_{i,t}^2 + \frac{\sigma^2}{4}\right)dt + \sum_{i=1}^{n}\sigma X_{i,t}dw_t$$

$$= \left(-\kappa\sum_{i=1}^{n} X_{i,t}^2 + \frac{n\sigma^2}{4}\right)dt + \sigma\sum_{i=1}^{n} X_{i,t}dw_t$$

$$= \left(-\kappa r_t + \frac{n\sigma^2}{4}\right)dt + \sigma\sqrt{r_t}\frac{\sum_{i=1}^{n} X_{i,t}}{\sqrt{r_t}}dw_t$$

$$= \left(-\kappa r_t + \frac{n\sigma^2}{4}\right)dt + \sigma\sqrt{r_t}dw_t'$$

$dw_t' = \frac{\sum X_{i,t}}{\sqrt{r_t}} dw_t$. E(dw_t') = 0. Quadratic variation of dw_u' over [0,t] is $\int_0^t (dw_u')^2$ $= \int_0^t \left(\sum_{i=1}^{n} X_{i,u}/\sqrt{r_u}\right)^2 du = \int_0^t \sum_{i=1}^{n} X_{i,u}^2/r_u \, du = t$. Hence w_t' is Wiener N(0,t). $\frac{n\sigma^2}{4} = \kappa\theta$

Hence degrees of freedom of chi-square, n = $4\kappa\theta/\sigma^2$

01 Diffusion Process Estimation

Cox-Ingersoll-Ross Process

Convert the $X_{i,t}$'s to r.v.s with unit variances.

$$X'_{i,t} = \frac{X_{i,t}}{\sqrt{\frac{\sigma^2}{4\kappa}\left(1 - e^{-\kappa(t-s)}\right)}}$$

$$E\left[X'_{i,t} \mid X'_{i,s}\right] = \frac{e^{-\kappa(t-s)/2} X_{i,s}}{\sqrt{\frac{\sigma^2}{4\kappa}\left(1 - e^{-\kappa(t-s)}\right)}}$$

$$V\left[X'_{i,t} \mid X'_{i,s}\right] = 1$$

$$r_t = X^2_{1,t} + X^2_{2,t} + X^2_{3,t} + \ldots + X^2_{n,t}$$

$$= \frac{\sigma^2}{4\kappa}\left(1 - e^{-\kappa(t-s)}\right)\left[X'^2_{1,t} + X'^2_{2,t} + X'^2_{3,t} + \ldots + X'^2_{n,t}\right]$$

$$\lambda = \sum_{i=1}^{n} E\left[X'_{i,t} \mid X'_{i,s}\right]^2 = \sum_{i=1}^{n} \frac{e^{-\kappa(t-s)} {X_{i,s}}^2}{\frac{\sigma^2}{4\kappa}\left(1 - e^{-\kappa(t-s)}\right)}$$

$$= \frac{e^{-\kappa(t-s)}}{\frac{\sigma^2}{4\kappa}\left(1 - e^{-\kappa(t-s)}\right)} \sum_{i=1}^{n} {X_{i,s}}^2$$

$$= \frac{4\kappa e^{-\kappa(t-s)}}{\sigma^2\left(1 - e^{-\kappa(t-s)}\right)} r_s$$

Conditional on $X_{i,0}$ at s = 0, sum of squares of means of standardized $X'_{i,s}$ is $\lambda = \frac{4\kappa e^{-\kappa t}}{\sigma^2(1-e^{-\kappa t})} r_0$ where λ is the non-centrality parameter of the chi-square distribution. Hence r_t is $\frac{\sigma^2(1-e^{-\kappa t})}{4\kappa} \times \sum_{i=1}^{n} X'^2_{i,t}$. The latter $\sum_{i=1}^{n} X'^2_{i,t}$ (sum of n squares of n.i.d. $X'_{i,t} \sim N(\mu_i, 1)$ is $\chi^2_{n,\lambda}$ with $n = 4\kappa\theta/\sigma^2$ and $\lambda = \frac{4\kappa e^{-\kappa t}}{\sigma^2(1-e^{-\kappa t})} r_0$.

01 Diffusion Process Estimation

Cox-Ingersoll-Ross Process

Probability density function of $\chi^2_{n,\lambda}$ is f (X'=x; n, λ) = $\frac{1}{2}e^{-\frac{x+\lambda}{2}}\left(\frac{x}{\lambda}\right)^{\frac{n-2}{4}} I_{\frac{n-2}{2}}\left(\sqrt{\lambda x}\right)$

Use the following transformations:

$$c = \frac{2\kappa}{\sigma^2\left(1-e^{-\kappa(t-s)}\right)}$$

$$u = ce^{-\kappa(t-s)}r_s$$

$$v = cr_t$$

$$q = \frac{2\kappa\theta}{\sigma^2} - 1$$

$$n = \frac{4\kappa\theta}{\sigma^2} = 2\left(\frac{2\kappa\theta}{\sigma^2} - 1\right) + 2 = 2q + 2$$

$$\lambda = \frac{4\kappa e^{-\kappa(t-s)}}{\sigma^2\left(1-e^{-\kappa(t-s)}\right)} r_s = 2u$$

$$r_t = \frac{\sigma^2}{4\kappa}\left(1-e^{-\kappa(t-s)}\right)x = \frac{1}{2c}x$$

Hence $r_t \sim \frac{1}{2c}\chi^2_{n,\lambda}$

r = 1/(2c) x or h(x). x = 2cr = 2v, or $h^{-1}(r)$.

To find pdf of r_t : g(r_t= r; n,λ) dr = [f(h^{-1}(r); n, λ) |dx/dr|] dr

Thus,

$$g(r; n,\lambda) = 2c\frac{1}{2}e^{-\frac{x+\lambda}{2}}\left(\frac{x}{\lambda}\right)^{\frac{n-2}{4}} I_{\frac{n-2}{2}}\left(\sqrt{\lambda x}\right)$$

$$= 2c\frac{1}{2}e^{-\frac{2v+2u}{2}}\left(\frac{2v}{2u}\right)^{\frac{2q+2-2}{4}} I_{\frac{2q+2-2}{2}}\left(\sqrt{2u2v}\right)$$

$$= ce^{-(v+u)}\left(\frac{v}{u}\right)^{\frac{q}{2}} I_q\left(2\sqrt{uv}\right)$$

02 Method of Moments and Infinitesimal Generator

When the likelihood function cannot be computed explicitly, we can use less efficient estimators obtained by optimizing an alternative criterion which admits an analytical form. A natural candidate for an optimization criterion arises from the moment conditions associated with a stochastic differential equation. We review various methods proposed in the literature in the context of unidimensional diffusion processes. This method works provided $\Phi(y)$ is stationary, μ, σ are functions of y, and $A\Phi$ converges.

The moment conditions studied by Hansen and Scheinkman (1995) are based on the following proposition:

PROPOSITION 12.1: *Let us consider a unidimensional process (y_t) that satisfies the stochastic differential equation*

$$dy_t = \mu(y_t)dt + \sigma(y_t)dW_t,$$

and let us introduce the infinitesimal generator A associated with this equation. It transforms a function Φ of y into

$$A\Phi(y) = \lim_{h\to 0} \frac{1}{h} E[(\Phi(y_{t+h}) - \Phi(y_t)) | y_t = y]$$

$$= \frac{d\Phi(y)}{dy}\mu(y) + \frac{1}{2}\frac{d^2\Phi(y)}{dy^2}\sigma^2(y).$$

Then, for a large set of functions Φ, $\tilde{\Phi}$, the following moment conditions are satisfied:

(i) $EA\Phi(y_t) = 0, \forall\Phi.$

(ii) $E[A\Phi(y_{t+1})\tilde{\Phi}(y_t) - \Phi(y_{t+1})A\tilde{\Phi}(y_t)] = 0, \forall\Phi,\tilde{\Phi}.$

02 Method of Moments and Infinitesimal Generator

By definition, the infinitesimal generator represents the infinitesimal drift of a transformed process. The differential expression of the generator follows from Ito's formula applied to $\Phi(y_t)$.

The moment conditions given in Proposition 12.1 concern the marginal moments of nonlinear functions of y_t (condition i) and cross moments of nonlinear functions of y_t and y_{t+1} (condition ii).

This set of moment conditions seems to be quite large since the functions Φ, $\tilde{\Phi}$ are not constrained (except for the second-order differentiability restriction). As an illustration, we detail the moment conditions for the exponential functions of y: $\Phi(y) = \exp(-ay)$, $\tilde{\Phi}(y) = \exp(-by)$. For condition (*i*), we get

$$E\left\{\exp(-ay_t)\left[\mu(y_t) - \frac{a}{2}\sigma^2(y_t)\right]\right\} = 0, \qquad \forall a.$$

for condition (*ii*),

$$E\left\{a\ \exp(-ay_{t+1} - by_t)\left[\mu(y_{t+1}) - \frac{a}{2}\sigma^2(y_{t+1})\right] - b\ \exp(-by_{t+1} - ay_t)\right.$$

$$\left.\left[\mu(y_t) - \frac{b}{2}\sigma^2(y_t)\right]\right\} = 0, \qquad \forall a,b.$$

02 Method of Moments and Infinitesimal Generator

For identification problem, consider the following diffusion model:

$$dy_t = (\alpha + \beta y_t)dt + \sigma y_t^{\gamma} dW_t,$$

which has been proposed by Chan et al. (1992) for the short-term interest rate. This specification encompasses a number of well-known continuous time models (see Broze, Scaillet, and Zakoian 1995). Its dynamics depends on four parameters: α, β, σ, and γ. The condition (i) for the exponential functions is

$$E\left[\exp(-ay_t)\left(\alpha + \beta y_t - \frac{a}{2}\sigma^2 y_t^{2\gamma}\right)\right] = 0, \qquad \forall a.$$

The parameters that are identifiable from conditions (i) are only γ, α/σ^2, and β/σ^2.

But if more conditions are provided, then σ can be separably identified.

02 Method of Moments and Infinitesimal Generator

Method of Moments

Consider the GMM implementation for restrictions in mean of infinitesimal generator. Select a priori *n* functions Φ_i , i = 1, ... ,*n*. The moment conditions are:

$$E\left[\frac{d\Phi_i}{dy}(y_t)\mu(y_t;\theta)+\frac{1}{2}\frac{d^2\Phi_i}{dy^2}(y_t)\sigma^2(y_t;\theta)\right]=0, \qquad i=1,\ldots,n, \quad (12.14)$$

Denote $\frac{d\Phi}{dy}(y_t)=\left[\frac{d\Phi_1}{dy}(y_t),\ldots,\frac{d\Phi_n}{dy}(y_t)\right]'$, and $\frac{d^2\Phi}{dy^2}(y_t)=\left[\frac{d^2\Phi_1}{dy^2}(y_t),\ldots,\frac{d^2\Phi_n}{dy^2}(y_t)\right]'$.

The moment estimator is derived as a solution of the optimization

$$\hat{\theta}_T=\underset{\theta}{\text{Arg min}}\sum_{t=1}^{T}\left\{\left[\mu(y_t;\theta)\frac{d\Phi'}{dy}(y_t)+\frac{1}{2}\sigma^2(y_t;\theta)\frac{d^2\Phi'}{dy^2}(y_t)\right]\right.$$
$$\left.\Omega\left[\mu(y_t;\theta)\frac{d\Phi}{dy}(y_t)+\frac{1}{2}\sigma^2(y_t;\theta)\frac{d^2\Phi}{dy^2}(y_t)\right]\right\}, \quad (12.15)$$

where Ω is a positive definite matrix of weights. When this matrix is an identity matrix, optimization yields

$$\hat{\theta}_T=\underset{\theta}{\text{Arg min}}\sum_{i=1}^{n}\left[\sum_{t=1}^{T}\mu(y_t;\theta)\frac{d\Phi_i}{dy}(y_t)+\frac{1}{2}\sigma^2(y_t;\theta)\frac{d^2\Phi_i}{dy^2}(y_t)\right]^2. \quad (12.16)$$

02 Method of Moments and Infinitesimal Generator

Method of Moments

EXAMPLE 12.1: Let us consider the model of Chan et al. with the identifying constraint $\alpha = 1$. The optimization criterion is

$$\sum_{i=1}^{n}\left[\sum_{t=1}^{T}(\exp - a_i y_t)\left(1 + \beta y_t - \frac{a_i}{2}\sigma^2 y_t^{2\gamma}\right)\right]^2,$$

where $a_1, \ldots, a_n$ are given real numbers. This is a quadratic objective with respect to the parameters β and σ^2 and hence it may first be concentrated (i.e., optimized) with respect to these parameters. The concentrated objective function depends only on the parameter γ, which may be numerically found by a grid search method.

03 Simulation Methods

Method of Simulated Moments

Contrary to the approach based on the infinitesimal generator, the method of simulated moments (MSM) does not require moment conditions in explicit form (see Duffie and Singleton 1993 for a particular application). We select a priori some functions $g_i(y_t, y_{t-1}, \ldots, y_{t-p})$, $i = 1, \ldots, n$, that depend on the present and p lagged values of the process. If this process satisfies the stochastic differential equation

$$dy_t = \mu(y_t;\theta)dt + \sigma(y_t;\theta)dW_t, \tag{12.31}$$

and is stationary, then the expectation

$$Eg_i(y_t, y_{t-1}, \ldots, y_{t-p}) = m_i(\theta), \qquad i = 1, \ldots, n, \tag{12.32}$$

is a function of the parameters. In general, m_i has a complicated form; therefore, the standard moment estimator (Hansen 1982)

$$\hat{\theta}_T = \operatorname{Arg}\min_{\theta} \sum_{t=1}^{T} [g(y_t, y_{t-1}, \ldots, y_{t-p}) - m(\theta)]' \Omega \sum_{t=1}^{T} [g(y_t, y_{t-1}, \ldots, y_{t-p}) - m(\theta)],$$

where Ω is a given definite symmetric matrix, cannot be computed even numerically.

03 Simulation Methods

Method of Simulated Moments

The principle of the method of simulated moments consists of replacing in the last criterion the unknown function m by an approximation obtained from simulations. For this purpose, we require simulated values $y_t^s(\theta)$, $t = 1, \ldots, ST$, compatible with the diffusion model (12.31) and the value θ of the parameter (see Chapter 11, Section 11.3.3, for the description of the simulation approach). A good approximation of $m(\theta)$ is

$$\hat{m}_T^s(\theta) = \frac{1}{ST} \sum_{t=1}^{ST} g[y_t^s(\theta), \ldots, y_{t-p}^s(\theta)], \tag{12.33}$$

where ST is sufficiently large. Then, the MSM estimator is defined by

$$\hat{\theta}_T^s = \underset{\theta}{\text{Arg min}} \sum_{t=1}^{T} [g(y_t, y_{t-1}, \ldots, y_{t-p}) - \hat{m}_T^s(\theta)]' \Omega \sum_{t=1}^{T} [g(y_t, y_{t-1}, \ldots, y_{t-p}) - \hat{m}_T^s(\theta)],$$

or, equivalently, by

$$\hat{\theta}_T^s = \underset{\theta}{\text{Arg min}} [\hat{m}_T - \hat{m}_T^s(\theta)]' \Omega [\hat{m}_T - \hat{m}_T^s(\theta)], \tag{12.34}$$

where $\hat{m} = \frac{1}{T} \Sigma_{t=1}^{T} \; g(y_t, y_{t-1}, \ldots, y_{t-p})$.

This method is essentially a calibration approach that provides the best possible match between the empirical moments based on the observations and those based on the simulated values.

This approach requires only the possibility of generating artificial observations on the model. Therefore, it can also be used in more general cases, such as multivariate diffusion models, continuous time models with stochastic volatility, or models that include jump components

03 Simulation Methods

Method of Indirect Inference

Suppose that we are interested in estimating the parameter θ, but the log-likelihood function $\Sigma_{t=1}^{T} \log f(y_t|\underline{y_{t-1}};\theta) = \Sigma_{t=1}^{T} \log f_t(\theta)$ (say) cannot be written in an explicit form. Empirically, a common approach consists of replacing the likelihood function by a simpler expression $\Sigma_{t=1}^{T} \log g(y_t|\underline{y_{t-1}};\theta) = \Sigma_{t=1}^{T} \log g_t(\theta)$, say, and in maximizing this modified log-likelihood function. Obviously, the replacement of f by g induces a specification error and likely results in inconsistent approximations of θ. Indirect inference is a simulation-based technique designed to correct for the asymptotic bias (see Gourieroux, Monfort, and Renault 1993; Gallant and Tauchen 1996). It requires two inputs: (1) simulated data compatible with the true model and the value θ of the parameter and (2) an auxiliary model (called an *instrumental model*) (Dhaene, Gourieroux, and Scaillet 1998) with a log-likelihood function $\Sigma_{t=1}^{T} \log g_t(\beta)$ that is easy to optimize numerically.

03 Simulation Methods

Method of Indirect Inference

Indirect inference includes the following steps:

1. The estimation of the auxiliary parameter β from the instrumental model and the observations:

$$\hat{\beta}_T = \underset{\beta}{\text{Arg max}} \sum_{t=1}^{T} \log g_t(\beta).$$

2. The simulation of the true model for the value θ of the parameter, yielding an artificial series of length ST: $y_t^s(\theta)$, $s = 1, \ldots, ST$, and then the estimation of the auxiliary parameter from the instrumental model and the artificial data:

$$\hat{\beta}_{TS}^s(\theta) = \underset{\beta}{\text{Arg max}} \sum_{t=1}^{ST} \log g[y_t^s(\theta) \,|\, \underline{y_{t-1}^s}(\theta); \beta].$$

3. The last step is a calibration step, yielding the indirect inference estimator of θ defined by

$$\hat{\theta}_T^s = \underset{\theta}{\text{Arg min}} \, [\hat{\beta}_T - \hat{\beta}_{TS}^s(\theta)]' \Omega [\hat{\beta}_T - \hat{\beta}_{TS}^s(\theta)]. \qquad (12.35)$$

where Ω is a weighting matrix.

03 Simulation Methods

Efficient Method of Moments

An alternative approach has been introduced by Gallant and Tauchen (1996). It is based on an auxiliary model $g(y_t|\underline{y_{t-1}};\beta)$ and makes use of the quasi-maximum likelihood (QML) estimator $\hat{\beta}_T$. This estimator satisfies the first-order condition

$$\frac{1}{T}\sum_{t=1}^{T}\frac{\partial \log g_t(\hat{\beta}_T)}{\partial\beta}=0.$$

It converges asymptotically to the solution β_0^* of the limiting moment condition

$$E\left[\frac{\partial \log g(y_t|\underline{y_{t-1}};\beta_0^*}{\partial\beta}\right]=0, \qquad (12.36)$$

called the *pseudo-true value*. The moment condition of (12.36) resembles a simulated method of moments condition. More precisely, Gallant and Tauchen consider the series $y_t^s(\theta)$, $s=1,\ldots,ST$, generated under the true model and find the solution of the following optimization:

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Efficient Method of Moments

$$\hat{\hat{\theta}}^s = \operatorname{Arg}_{\theta} \min \left[\frac{1}{ST} \sum_{t=1}^{T} \sum_{s=1}^{S} \frac{\partial \log g(y_t^s(\theta) \,|\, \underline{y_{t-1}^s}; \hat{\beta}_T)}{\partial \beta} \right]$$

$$\Omega \left[\frac{1}{ST} \sum_{t=1}^{T} \sum_{s=1}^{S} \frac{\partial \log g(y_t^s(\theta) \,|\, \underline{y_{t-1}^s}; \hat{\beta}_T)}{\partial \beta} \right], \tag{12.37}$$

where Ω is a weighting matrix. It has been proved that the estimators $\hat{\hat{\theta}}^s$ and $\hat{\theta}^s$ given in (12.37) and (12.35) are asymptotically equivalent. This is a consequence of the interpretation of the objective function as a Wald test statistic [see (12.35)] and a Lagrange multiplier test statistic [see (12.37)], respectively. The efficiency of the estimator increases with the number of moments considered. In the limiting case of an auxiliary model with an infinite number of parameters, full efficiency is achieved. This explains the term *efficient method of moments* (EMM) used as a reference in the literature (Gallant and Tauchen 1996).

03 Simulation Methods

APPLICATION EXAMPLE

Let us consider a continuous time process satisfying a stochastic differential equation

$$dy_t = \mu(y_t;\theta)dt + \sigma(y_t;\theta)dW_t, \tag{12.38}$$

where (W_t) is a standard Brownian motion. When the available observations are sampled at integer dates $1, 2, \ldots, T$, it is generally impossible to determine the analytical form of the likelihood function. A common approach consists of replacing the initial continuous time model in (12.25) by its Euler discretization (see Section 11.3.1):

$$y_t = y_{t-1} + \mu(y_{t-1};\beta) + \sigma(y_{t-1},\beta)\varepsilon_t, \tag{12.39}$$

where (ε_t) is a Gaussian white noise. Next, we estimate β by the maximum likelihood method applied to the approximated model in (12.26) and likely conclude that the estimated parameter $\hat{\beta}$ provides a good approximation of the unknown θ. However, since the Euler discretization is an approximation, model (12.39) is misspecified, causing an asymptotic bias of its estimators, which may be arbitrarily large (see Section 12.5.4).

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APPLICATION EXAMPLE

However, the Euler discretization can be used as an instrumental model in the indirect inference method. The idea is to introduce a second Euler discretization involving a very short time unit, $\delta = 1/10$ (say). More precisely, we define the process (y_t^{δ}, $t = k\delta$, k varying) such that

$$y_{(k+1)\delta}^{(\delta)} = y_{k\delta}^{(\delta)} + \delta\mu\left(y_{k\delta}^{(\delta)}, \theta\right) + \sigma\left(y_{k\delta}^{(\delta)}, \theta\right)\sqrt{\delta}\varepsilon_k^{(\delta)}, \tag{12.40}$$

where $\varepsilon_k^{(\delta)}$, k varying, is a Gaussian white noise. This finer Euler discretization can be used to simulate the continuous time process. Let us denote by $y_{k\delta}^{(\delta),s}(\theta)$, $k = 1, \ldots, T/\delta$, the simulated path corresponding to the parameter value θ, and $y_t^{(s)}(\theta) = y_t^{(\delta),s}(\theta)$, $t = 1, \ldots, T$, the values corresponding to the observation dates.

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APPLICATION EXAMPLE

The indirect inference approach includes the three estimation steps outlined below:

1. Estimation of β from observations

$$\hat{\beta}_T = \underset{\beta}{\text{Arg min}} \sum_{t=1}^{T} \left\{ -\frac{1}{2} \log \sigma^2(y_{t-1},\beta) - \frac{1}{2} \frac{[y_t - \mu(y_{t-1},\beta)]^2}{\sigma^2(y_{t-1};\beta)} \right\};$$

2. Estimation of β from simulations

$$\hat{\beta}_T^s(\theta) = \underset{\beta}{\text{Arg min}} \sum_{t=1}^{T} \left\{ -\frac{1}{2} \log \sigma^2(y_{t-1}^{(s)}(\theta),\beta) - \frac{1}{2} \frac{[y_t^{(s)}(\theta) - \mu(y_{t-1}^{(s)}(\theta),\beta)]^2}{\sigma^2(y_{t-1}^{(s)}(\theta);\beta)} \right\};$$

3. Calibration

$$\hat{\theta}_T^s = \underset{\beta}{\text{Arg min}} \, [\hat{\beta}_T - \hat{\beta}_T^s(\theta)]' \Omega [\hat{\beta}_T - \hat{\beta}_T^s(\theta)],$$

where Ω is a given matrix of weights.

04 Jump Diffusion Process

Ref: "Estimation of Jump-diffusion processes based on Indirect inference," by George J. Jiang, IFAC Computation in Economics, Finance and Engineering Economic Systems, Cambridge 1998.

$$\frac{dS_t}{S_t} = (\alpha_t(\beta) - \lambda\mu_0)dt + \sigma_t(\beta)dW_t + (Y_t(\beta) - 1)dq_t(\lambda) \quad (1)$$

Alternatively, (1) can be rewritten in terms of the logarithmic asset prices, i.e., $s_t = ln S_t$, as:

$$ds_t = \mu_t(\beta)dt + \sigma_t(\beta)dW_t + \ln Y_t(\beta)dq_t(\lambda) \quad (2)$$

where $\mu_t = \alpha_t - \lambda\mu_0 - \frac{1}{2}\sigma_t^2$.

Special cases of the model defined in (2) include those by Press (1967) with $\sigma_t = \sigma, \mu_t = 0$, Merton (1976) with $\sigma_t = \sigma, \mu_t = \mu$, Beckers (1981) and Ball and Torous (1985) with $\sigma_t = \sigma, \mu_t = \mu, \mu_0 = 0$, and Lo (1988) with $\ln Y_t = \kappa(s_t)$, i.e. the jump size is determined by the process itself.

S_t — asset price at time t;
α_t — the instantaneous expected return;
σ_t — the instantaneous volatility of the asset's return conditional on that the Poisson jump event does not occur;
W_t — a standard Gauss-Wiener process or the Brownian motion process;
$q_t(\lambda)$ — a Poisson process which is iid over time;
λ — the intensity parameter of the Poisson distribution;
$Y_t - 1$ — the random jump size with $Y_t \geq 0$;
μ_0 — expectation of the relative jump size, i.e. $\mu_0 = E[Y_t - 1]$;
$dq_t(\lambda), dW_t$ — statistically independent;

$\theta = (\beta, \mu_0, \lambda) \in \Theta$ — the parameter space which parameterizes the coefficient functions, the jump sizes, as well as the intensity of Poisson process.

04 Jump Diffusion Process

Estimating exchange rate processes of Pound, Mark, Yen, Franc before euro was introduced in 1999.

Model I: *Black-Scholes Non-Jump Model*

$$ds_t = \mu dt + \sigma dW_t \tag{5}$$

Model II: *Merton's Jump Model*

$$ds_t = (\mu - \lambda\mu_0)dt + \sigma dW_t + \ln Y_t dq_t(\lambda) \tag{6}$$

Model III: *Conditional Heteroscedasticity and Jump*

$$ds_t = (\mu - \lambda\mu_0)dt + (\sigma + \sigma_1 s_t)dW_t + \ln Y_t dq_t(\lambda) \tag{7}$$

Model IV: *Mean-Reversion, Conditional Heteroscedasticity and Jump*

$$ds_t = (\mu - \beta s_t - \mu_0\lambda)dt + (\sigma + \sigma_1 s_t)dW_t + \ln Y_t dq_t(\lambda) \tag{8}$$

where $\ln Y_t \sim \text{iid } N(\mu_0, v^2)$.

04 Jump Diffusion Process

Estimating exchange rates processes of Pound, Mark, Yen, Franc by Indirect Inference

Step 1: Choice of the instrumental or auxiliary model: For the continuous-time jump-diffusion process (2), a natural choice of the instrumental or auxiliary model is its simple discretization. However, when $\lambda\Delta_i$ is small (< 1) as in our application, a much simpler instrumental or auxiliary model is given by

$$s_{i+1} = s_i + \mu(t_i;\beta_I)\Delta_i + \sigma(t;\beta_I)\Delta_i^{1/2}\epsilon_{i+1}^0 + \eta(\mu_{0I} + \nu_I\epsilon_{i+1}^1) \tag{10}$$

with instrumental parameter $\theta_I = (\beta_I, \mu_{0I}, \nu_I, \lambda_I) \in \Theta_I$ – the parameter space, where $s_i = s_{t_i}, \forall i$, $\Delta_i = t_{i+1} - t_i$, $\epsilon_{i+1}^j \sim$ iid $N(0,1)$, $j = 0, 1$; $i = 0, 1, \ldots, M-1$, and $\eta \sim$ Bernoulli distribution with $P(\eta = 1) = \lambda_I\Delta_i$ and $P(\eta = 0) = 1 - \lambda_I\Delta_i$. The model defined in (10) has a one-to-one relationship with (2), and the parameter space Θ_I has the same dimension as Θ, i.e. $dim(\Theta_I) = dim(\Theta)$. Both models are Markov processes in continuous state space, with (2) being in continuous-time while (10) in discrete-time.

The conditional density function of s_{i+1} given s_i for the instrumental model can be written as

$$f_I(s_{i+1}|s_i;\theta_I) = (1 - \lambda_I\Delta_i)\phi_1(s_{i+1}) + \lambda_I\Delta_i\phi_2(s_{i+1}) \tag{11}$$

where $\phi_1(\cdot)$ is the pdf of the normal distribution with mean $s_i + \mu(t_i;\beta_I)\Delta_i$ and variance $\sigma(t;\beta_I)^2\Delta_i$ and $\phi_2(\cdot)$ is the pdf of the normal distribution with mean $s_i + \mu(t_i;\beta_I)\Delta_i + \mu_{0I}$ and variance $\sigma(t;\beta_I)^2\Delta_i + \nu_I^2$.

Step 2: Estimation of the instrumental or auxiliary model: The ML estimator of θ_I is given by

$$\hat{\theta}_I = \arg\max_{\theta_I} \sum_{i=0}^{M-1} \ln f_I(s_{i+1}|s_i;\theta_I) \tag{12}$$

which is much easier to implement than the Poisson driven jump-diffusion process.

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Step 3: Path simulation of the original jump-diffusion process and estimation of the instrumental model based on simulated sampling path: Since the instrumental or auxiliary model is the discretized approximation of the original one, i.e. (M_I) and $f_I(\cdot)$ are misspecified, the pseudo maximum likelihood (PML) estimator $\hat{\theta}_I$ is generally biased and inconsistent estimator of the true parameter θ. The indirect inference estimation method uses simulations performed under the original model to correct for the asymptotic biases of $\hat{\theta}_I$. Given values of θ and initial values of s_t at $t = t_0$, we can simulate the sampling path $\tilde{s}_t$ of s_t, observed at $(t_0, t_1, \ldots, t_M)$, for the original jump-diffusion model (2). As in most simulation exercises, we can redraw such simulations H times. Then we estimate the parameter θ_I of the instrumental model, denoted by $\hat{\theta}_I^{HM}(\theta)$, from the observations of the simulated sampling path via ML method

$$\hat{\theta}_I^{HM}(\theta) = \arg\max_{\theta_I} \sum_{h=1}^{H} \sum_{i=0}^{M-1} \ln f_I(\tilde{s}_{i+1}^h(\theta) | \tilde{s}_i^h(\theta); \theta_I) \tag{13}$$

Simulation of the exact dynamic sampling path of the jump-diffusion model defined in (2) is in general impossible unless its transition density functions are explicitly known whereas in this case the model could be estimated directly. Hence in general we can only simulate approximated sampling path of the jump-diffusion model. Discrete approximations of the continuous-time jump-diffusion process can be obtained through the Euler scheme. Alternatively, the Milshtein scheme can be used for the continuous part of the process as it has better convergence rate than the Euler scheme for the convergence in $L^p(\Omega)$ and the almost sure convergence, see (Talay, 1996).

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Step 4: Indirect estimator of jump-diffusion parameter: An indirect estimator of θ based on M observations of the simulated sampling path with H drawings, denoted by $\hat{\theta}^{HM}$, is defined by choosing values of θ from which $\hat{\theta}_I$ and $\hat{\theta}_I^{HM}(\theta)$ are as close as possible, i.e.

$$\hat{\theta}^{HM}(\Omega) = \arg\min_{\theta}(\hat{\theta}_I - \hat{\theta}_I^{HM}(\theta)) \\ \Omega(\hat{\theta}_I - \hat{\theta}_I^{HM}(\theta))' \qquad (14)$$

where Ω is a symmetric nonnegative matrix, defining the metric or the weighting scale. The functions $\hat{\theta}_I^{HM}(\theta)$ relate the parameters of the original model to the estimated parameter values of the instrumental model.

Gouriéroux, Monfort and Renault (1993) show that, under certain regularity conditions, the indirect inference estimator $\hat{\theta}^{HM}(\Omega)$ is consistent, asymptotically normal, when H is fixed and M goes to infinity:

$$\sqrt{M}(\hat{\theta}^{HM}(\Omega^*) - \theta) \xrightarrow{d} N(0, V(H, \Omega^*)) \quad (16)$$

where $V(H, \Omega^*) = (1 + \frac{1}{H})[\frac{\partial b'(\theta_0)}{\partial \theta}\Omega^* \frac{\partial b(\theta_0)}{\partial \theta}]^{-1}$, and Ω^* is the optimal weighting matrix. The choice of Ω in the above application is the identity matrix as the dimensions of the parameter space are equal for the original model and the instrumental model, i.e. $dim(\Theta) = dim(\Theta_I)$.

04 Jump Diffusion Process

Estimating exchange rates processes of Pound, Mark, Yen, Franc by Indirect Inference

The data covers the period from January 7, 1976 to January 17, 1996 with 1,046 weekly observations.

(a) UK pound/US dollar

	Diffusion		Jump		
	$\beta(10^{-2})$	$\sigma_1(10^{-3})$	λ	$\mu_0(10^{-4})$	$\nu(10^{-3})$
II			2.70	−12.5	8.90
			(0.38)	(2.59)	(0.64)
III		−13.2	1.71	−11.7	9.25
		(2.70)	(0.28)	(3.18)	(0.73)
IV	4.02	−13.3	1.72	−12.7	9.21
	(11.9)	(3.04)	(0.31)	(4.52)	(0.86)

(b) German mark/US dollar

II			2.94	−0.45	8.59
			(0.34)	(2.21)	(0.53)
III		−1.93	2.41	−0.73	8.64
		(1.01)	(0.25)	(1.96)	(0.50)
IV	3.71	−1.64	2.41	−0.32	8.63
	(8.74)	(0.95)	(0.26)	(2.15)	(0.50)

(c) Japanese yen/US dollar

	Diffusion		Jump		
	$\beta(10^{-2})$	$\sigma_1(10^{-3})$	λ	$\mu_0(10^{-4})$	$\nu(10^{-3})$
II			1.47	−16.8	11.0
			(0.27)	(5.74)	(0.93)
III		−0.50	0.96	−16.7	11.6
		(1.40)	(0.16)	(5.83)	(0.87)
IV	6.41	−0.51	0.98	−18.3	11.5
	(5.44)	(1.14)	(0.17)	(5.88)	(0.85)

(d) French franc/US dollar

II			2.81	4.39	8.58
			(0.38)	(2.49)	(0.60)
III		−21.3	1.73	4.57	8.83
		(2.47)	(0.19)	(2.31)	(0.62)
IV	8.72	−21.2	1.74	5.77	8.79
	(10.9)	(2.55)	(0.21)	(2.78)	(0.62)

Note: The numbers in the brackets are standard derivations of the above estimates.

The estimation results suggest strong evidence for the misspecification of both the non-jump Black-Scholes model and Merton's constant conditional volatility jump model as representation of currency exchange rate processes. In particular. models assuming conditional homoscedasticity tends to overestimate the jump frequency but underestimate the jump size.